Warehouse Performance Analysis Using T-Test, ANOVA and Time Series

A Mini Project Report submitted to

MOHAN BABU UNIVERSITY

in Partial Fulfillment of the Requirements for the Award of the degree of

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

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SCHOOL OF COMPUTING

MOHAN BABU UNIVERSITY

Sree Sainath Nagar, Tirupati – 517 102, A.P., INDIA

2026-27

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DEPARTMENT OF DATA SCIENCE

CERTIFICATE

This is to certify that the mini project report entitled

"Warehouse Performance Analysis Using T-Test, ANOVA and Time Series"

is the Bonafide work done by

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in the Department of **Data Science**, and submitted to Mohan Babu University, Tirupati in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering (Data Science) during the academic year 2026-2027. This work has beencarried out under my supervision. The results of this mini project work havenot been submitted to any university for the award of any degree or diploma.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

DEPARTMENT OF DATA SCIENCE

Vision

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Mission

- To impart quality education in Computer Science and Engineering with specializations in Data Science by disseminating knowledge through contemporary curriculum, competent faculty and effective teaching-learning methodologies.
- Nurture research, innovation and entrepreneurial skills among students and faculty to contribute to the needs of industry and society.
- Inculcate professional attitude, ethical and social responsibilities for prospective and promising Engineering profession.
- Encourage students to engage in life-long learning by creating awareness of the contemporary developments in Computer Science and Engineering with specialization in Data Science.

PROGRAM EDUCATIONAL OBJECTIVES

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PEO2. Become successful entrepreneurs or be employed by acquiring required skill sets in the domains of Data Science and allied areas.

PEO3. Exhibit progression and effective adaptation to technological developments through life-long learning to address ever changing industrial requirements and follow ethical attitude in professional practice.

PROGRAM SPECIFIC OUTCOMES

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PSO2. Develop intelligent systems using novel Machine Learning and Artificial Intelligence techniques.

PSO3. Design and develop efficient software systems using modern tools, techniques, and platforms to meet societal needs.

PSO4. Apply suitable tools and techniques to build secure distributed systems.

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On successful completion of the Program, the graduates of B.Tech. CSE (DS) Program will be able to:

PO1. **Engineering knowledge**: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2. **Problem analysis**: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3. **Design/development of solutions**: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4. **Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5. **Modern tool usage**: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6. **The engineer and society**: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

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PO8. **Ethics**: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

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PO10. **Communication**: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11. **Project management and finance**: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12. **Life-long learning**: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

DECLARATION

We hereby declare that this project report titled **"Warehouse Performance Analysis Using T-Test, ANOVA and Time Series"** is agenuine work carried out by us, in **B.Tech *(Computer Science and Engineering (Data Science))*** degree course of **Mohan Babu University, Tirupati** and has not been submitted to any other course or University for the award of any degree by us.

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea / data / fact / source in our submission. We understand that any violation of the above will cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Signature of the students

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ABSTRACT

In the modern supply chain and logistics industry, efficient warehouse management is essential for reducing operational costs, improving productivity, and ensuring timely order fulfillment. With the increasing volume of inventory and warehouse operations, analyzing performance accurately has become a challenging task. Traditional warehouse evaluation methods often rely on basic performance metrics and manual analysis, which may not provide sufficient insights into operational efficiency. This project presents a data-driven approach for Warehouse Performance Analysis using statistical techniques such as T-Test, ANOVA, and Time Series Analysis.The proposed system analyzes historical warehouse data containing important operational parameters such as order processing time, picking time, packing time, delivery time, inventory levels, order quantities, and employee productivity. Data preprocessing techniques are applied to handle missing values, remove inconsistencies, and prepare the dataset for statistical analysis.The T-Test is used to compare the performance of two groups, such as two warehouse shifts, employees, or operational periods, and determine whether there is a statistically significant difference between their performance. ANOVA (Analysis of Variance) is applied to compare the performance of multiple groups, such as different warehouse locations, departments, shifts, or delivery categories. These statistical techniques help identify significant factors affecting warehouse efficiency.Furthermore, Time Series Analysis is used to study performance trends over time. It helps identify patterns in order volume, processing time, inventory levels, and productivity and can be used to forecast future warehouse demand and performance. Data visualization techniques are also incorporated to present statistical results, trends, and key performance indicators through graphs and charts.The proposed system provides meaningful insights into warehouse operations and helps organizations identify performance variations, operational bottlenecks, and improvement opportunities. By combining T-Test, ANOVA, and Time Series Analysis, the system provides a comprehensive analytical framework for improving warehouse efficiency, resource utilization, inventory management, and decision-making. The approach can be effectively applied in warehouses, distribution centers, logistics companies, and supply chain management systems.

Keywords: Warehouse Performance Analysis, T-Test, ANOVA, Time Series Analysis, Data Analytics, Statistical Analysis, Warehouse Management, Supply Chain Management, Performance Optimization.

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CHAPTER-1
INTRODUCTION

1.1 Introduction:

With the rapid growth of e-commerce, logistics, and supply chain management, efficient warehouse operations have become essential for ensuring timely order fulfillment, reducing operational costs, and improving customer satisfaction. However, identifying performance variations, evaluating operational efficiency, and predicting future warehouse trends remain significant challenges. Factors such as order processing time, inventory levels, shipping delays, labor productivity, and order fulfillment rates can directly affect overall warehouse performance. Therefore, developing a data-driven system for warehouse performance analysis has become important in the fields of data analytics, operations management, and supply chain optimization.

This project proposes a comprehensive approach for analyzing warehouse performance using statistical techniques and time-series analysis. The system utilizes the T-test to compare the performance of two groups and determine whether their differences are statistically significant. ANOVA (Analysis of Variance) is applied to compare performance across multiple warehouses, departments, or operational categories to identify significant variations. These statistical techniques help evaluate operational efficiency and support evidence-based decision-making.For effective performance monitoring, time-series analysis is used to examine historical warehouse data and identify trends, seasonal patterns, and changes in operational performance over time.

The project aims to deliver a reliable, efficient, and user-friendly analytical system that assists warehouse managers, logistics professionals, and supply chain analysts in monitoring performance and identifying areas for improvement. By combining statistical hypothesis testing with time-series analysis, the system supports data-driven decisions, improves resource utilization, reduces operational inefficiencies, and contributes to better warehouse management.



Fig-1: Warehouse Performance Analysis Workflow

Fig 1. Warehouse Performance Analysis

In Fig-1. the figure represents the workflow of warehouse performance analysis, including data collection, preprocessing, statistical analysis, and time-series analysis. It highlights key visualizations and evaluation methods used to identify trends, compare performance, and improve warehouse efficiency.

1.2 Problem Statement:

1. With the increasing use of digital technologies in logistics, e-commerce, and supply chain management, warehouse performance analysis has become essential.
2. Warehouses handle large volumes of orders, inventory, and deliveries, making it challenging to maintain operational efficiency and timely order fulfillment.
3. Poor warehouse performance can lead to delayed shipments, increased operational costs, inventory management issues, and reduced productivity.
4. Without proper data analysis, identifying performance variations and predicting future operational trends becomes difficult.
5. An automated, efficient, and data-driven system is required to analyze key performance indicators such as order processing time, inventory turnover, delivery delays, and order fulfillment rates
6. The proposed system uses the *T-test* to compare two groups and determine whether their performance differences are statistically significant.
7. *ANOVA (Analysis of Variance)* is used to compare performance across multiple warehouses or operational categories.

1.3 Objectives:

- To develop a data-driven system for analyzing warehouse performance and improving operational efficiency.
- To compare the performance of two groups using the T-test and determine whether their differences are statistically significant.
- To apply ANOVA to compare performance across multiple warehouses, departments, or operational categories.
- To analyze historical warehouse data using time-series techniques to identify trends, seasonal patterns, and changes in performance over time.
- To preprocess and analyze warehouse datasets using data cleaning, statistical methods, and visualization techniques to obtain meaningful insights.
- To evaluate key performance indicators such as order processing time, delivery delays, inventory turnover, and order fulfillment rates.
- To provide a foundation for future enhancements, such as predictive analytics, real-time warehouse monitoring, demand forecasting, and advanced optimization techniques to improve warehouse productivity and decision-making.

1.4 Limitations:

- The system may be limited by the availability and quality of warehouse datasets, which can affect the accuracy of performance analysis.
- Missing, incorrect, or inconsistent warehouse records may lead to unreliable statistical results and misleading conclusions.
- The T-test and ANOVA results depend on assumptions such as data distribution, independence, and variance, which may not always be satisfied by real-world warehouse data.
- The system may not provide real-time monitoring if warehouse data is updated manually or collected at irregular intervals.
- The current system may not incorporate advanced predictive models or real-time IoT integration, which could improve forecasting accuracy and support faster operational decision-making.

CHAPTER-2
METHODOLOGY

The system is designed to systematically analyze warehouse performance and provide meaningful insights through a structured analytical workflow. The overall process includes data collection, data preprocessing, statistical analysis, time-series analysis, visualization, and performance evaluation. Initially, the system collects warehouse data from CSV or Excel files, including details such as order processing time, inventory levels, delivery delays, order fulfillment rates, and operational productivity. This flexible input approach allows the system to analyze warehouse performance across different operational conditions.Once the data is collected, it undergoes preprocessing using Python libraries such as Pandas and NumPy. This stage involves handling missing values, removing duplicate records, correcting data inconsistencies, and converting date columns into appropriate formats to improve data quality and ensure consistency for further analysis. The cleaned dataset is then used for statistical evaluation.The T-test is applied to compare the performance of two groups, such as two warehouses or two operational periods, to determine whether their mean performance differs significantly. Subsequently, ANOVA (Analysis of Variance) is used to compare the performance of three or more groups and identify statistically significant differences among multiple warehouses, departments, or operational categories. These statistical techniques help determine whether observed performance variations are meaningful or may have occurred by chance.Next, time-series analysis is performed on historical warehouse data to identify trends, seasonal patterns, and changes in operational performance over time. Performance indicators such as daily order volumes, processing times, and delivery delays are analyzed to understand historical behavior and, where appropriate, forecast future trends. Python libraries such as Matplotlib and Seaborn are used to visualize the results through charts, graphs, and performance comparisons.

Finally, the system presents the analytical results in a clear and user-friendly format, helping warehouse managers and supply chain professionals identify performance gaps, monitor operational efficiency, and make informed decisions. Overall, the workflow begins with warehouse data collection, followed by preprocessing, T-test analysis, ANOVA, time-series analysis, and visualization, forming a comprehensive data-driven warehouse performance analysis system.

CHAPTER-3

SYSTEM DESIGN

3.1 Requirements:

The system requires a computer with an Intel i3 or i5 processor or higher, with a multi-core processor recommended for faster data processing and statistical analysis. A minimum of 4 GB RAM is necessary, although 8 GB or more is recommended for smoother performance when handling large warehouse datasets. The system should have at least 2–5 GB of free storage space to accommodate datasets, Python libraries, analysis scripts, and output files. An internet connection may be required to install libraries, access cloud-based tools, or retrieve warehouse datasets. A monitor or display unit is needed to visualize statistical results, time-series trends, performance comparisons, charts, and graphs.

3.2 System Design:

The Warehouse Performance Analysis System is designed to provide an efficient and data-driven solution for evaluating warehouse operations and identifying performance trends. The system allows users to upload warehouse datasets in CSV or Excel format containing information such as order processing time, inventory levels, delivery delays, order fulfillment rates, and operational productivity. Time-series analysis is used to examine historical data, identify trends and seasonal patterns, and forecast future warehouse performance where appropriate.

1. Input Module:

Handles the collection of warehouse data for analysis.

Modes:

- CSV/Excel Upload: Allows users to upload warehouse datasets from local storage.
- Dataset Selection: Enables users to select datasets containing order processing time, inventory levels, delivery delays, order fulfillment rates, and warehouse productivity.

2. Data Preprocessing

The system prepares warehouse data for statistical analysis and time-series evaluation through several preprocessing operations. Missing values are handled, duplicate records are removed, and inconsistent data is corrected. Date columns are converted into appropriate formats, and relevant performance indicators are selected to ensure data quality and

consistency for accurate analysis.

3. Statistical Analysis

The statistical analysis module uses the T-test and ANOVA to evaluate differences in warehouse performance. The T-test compares the average performance of two groups, such as two warehouses or two time periods, while ANOVA compares the average performance of three or more groups. These techniques help determine whether observed differences are statistically significant and identify areas that may require operational improvement.

4. Time-Series Analysis and Visualization

The time-series analysis module examines warehouse data collected over days, weeks, or months to identify trends, seasonal variations, and changes in operational efficiency. It can analyze order volumes, processing times, and delivery delays to understand historical performance and forecast future trends when sufficient data is available.

5. Output Module:

Presents the final analytical results to the user.

Components:

Displays warehouse performance analysis results.

Shows T-test and ANOVA results to identify significant performance differences.

Displays time-series graphs showing trends and seasonal patterns in warehouse operations.

Provides key performance indicators, such as order processing time, delivery delays, and order fulfillment rates.

3.3 Technologies Used:

Technologies Used

1. Programming Language – Python: Used for data analysis and statistical calculations.
2. Data Processing – Pandas and NumPy: Used for data cleaning and manipulation.
3. Statistical Analysis – SciPy: Used to perform T-tests and ANOVA.
4. Time-Series Analysis – Statsmodels: Used to identify trends and forecast future performance.
5. Data Visualization – Matplotlib and Seaborn: Used to create graphs and charts.
6. Development Environment – Google Colab: Used to execute Python code and display results.
7. Dataset Format – CSV and Excel: Used to store and import warehouse data.

CHAPTER -4
IMPLEMENTATION

The implementation of the Warehouse Performance Analysis system involves collecting warehouse operational data, preprocessing the data, applying statistical techniques, and visualizing the results to evaluate warehouse efficiency and identify areas for improvement. The system uses Python, Pandas, NumPy, SciPy, Matplotlib, and Seaborn to perform data analysis and generate meaningful insights.

1. Data Collection: Warehouse operational data is collected from CSV or Excel files containing information such as order processing time, inventory levels, order quantities, picking and packing time, delivery delays, daily shipments, and warehouse productivity. The dataset may include multiple warehouses, different shifts, and daily or monthly records to support performance comparisons.

2. Data Preprocessing: The collected dataset is cleaned by handling missing values, removing duplicate records, correcting data types, and identifying abnormal values. Date columns are converted into the appropriate format, and numerical attributes are standardized where necessary. The data is then organized into suitable categories, such as warehouse location, shift, order processing duration, and shipment date, to ensure accurate analysis.

3. T-Test Analysis: The T-test is applied to compare the average performance of two groups, such as morning and evening shifts or two warehouse locations. For example, it can determine whether the average order processing time differs significantly between two shifts. The calculated test statistic and p-value help identify whether the observed difference is statistically significant at a selected significance level, commonly 0.05.

4. ANOVA Analysis: Analysis of Variance (ANOVA) is used to compare the average performance of three or more groups, such as multiple warehouses, shifts, or delivery teams. It helps determine whether significant differences exist in processing time, shipment volume, or productivity across the groups. When a significant difference is detected, suitable post-hoc tests can identify which specific groups differ.

5. Time-Series Analysis: Time-series techniques are used to analyze warehouse performance over time using daily, weekly, or monthly records. Metrics such as order volume, inventory movement, delivery delays, and processing time are examined to identify trends, seasonal patterns, and fluctuations. Moving averages and time-series forecasting can help estimate future workload, anticipate demand, and support better resource allocation.

6. Data Visualization and Reporting: The analysis results are presented through graphs, charts, and summary tables. Bar charts compare warehouse performance, line graphs display trends over time, and box plots show variations in processing time. The dashboard highlights key performance indicators, including average processing time, total orders processed, delivery delay rate, and productivity. These visualizations make the results easier to understand.

7. Performance Evaluation and Decision-Making: The final results are used to identify inefficient processes, compare warehouse operations, predict workload changes, and support operational planning. Based on the findings, warehouse managers can improve staffing schedules, optimize inventory handling, reduce processing delays, and increase overall productivity. The system provides a data-driven approach to warehouse performance monitoring and continuous improvement.

CHAPTER-5

PROPOSED SYSTEM ARCHITECTURE

The proposed Warehouse Performance Analysis system is designed as a modular and sequential architecture that integrates statistical analysis and time-series techniques to evaluate warehouse efficiency. Initially, warehouse operational data is collected from CSV or Excel files, including order processing time, inventory levels, shipment quantities, and delivery delays. Once the data is obtained, it is passed to the preprocessing module, where operations such as handling missing values, removing duplicates, and correcting data types are performed to ensure data quality and consistency.

The cleaned data is then forwarded to the statistical analysis module, where the T-test compares the average performance of two groups, while ANOVA identifies significant differences among three or more groups. The results help determine variations in warehouse productivity and operational efficiency. The data is subsequently passed to the time-series analysis module, which examines daily, weekly, or monthly performance trends and identifies patterns in order volume, processing time, and delivery delays. Finally, the analysis results are presented through charts and graphs, displaying key performance indicators and trends.

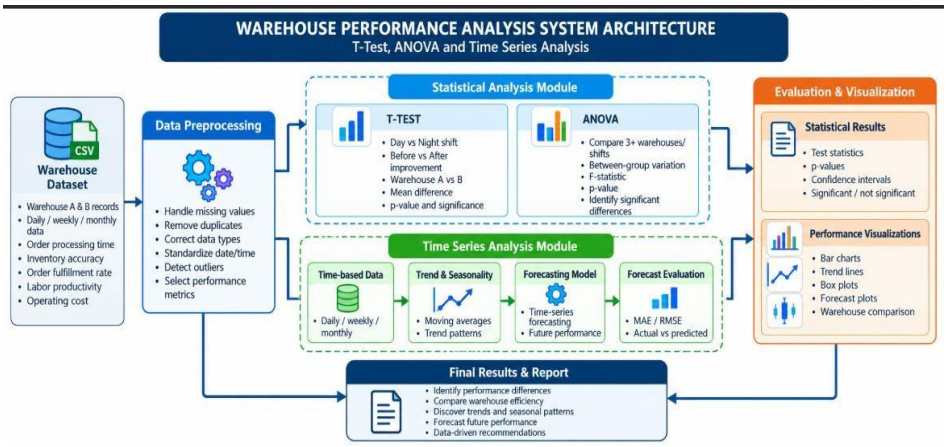


Fig 2: Warehouse Performance Analysis System Architecture

In Fig-2. The architecture illustrates the process of analyzing warehouse performance using statistical methods such as T-test and ANOVA, along with time-series analysis. It includes data preprocessing, performance evaluation, visualization, and report generation to identify trends and improve operational efficiency.

5.1 Flow Chart:

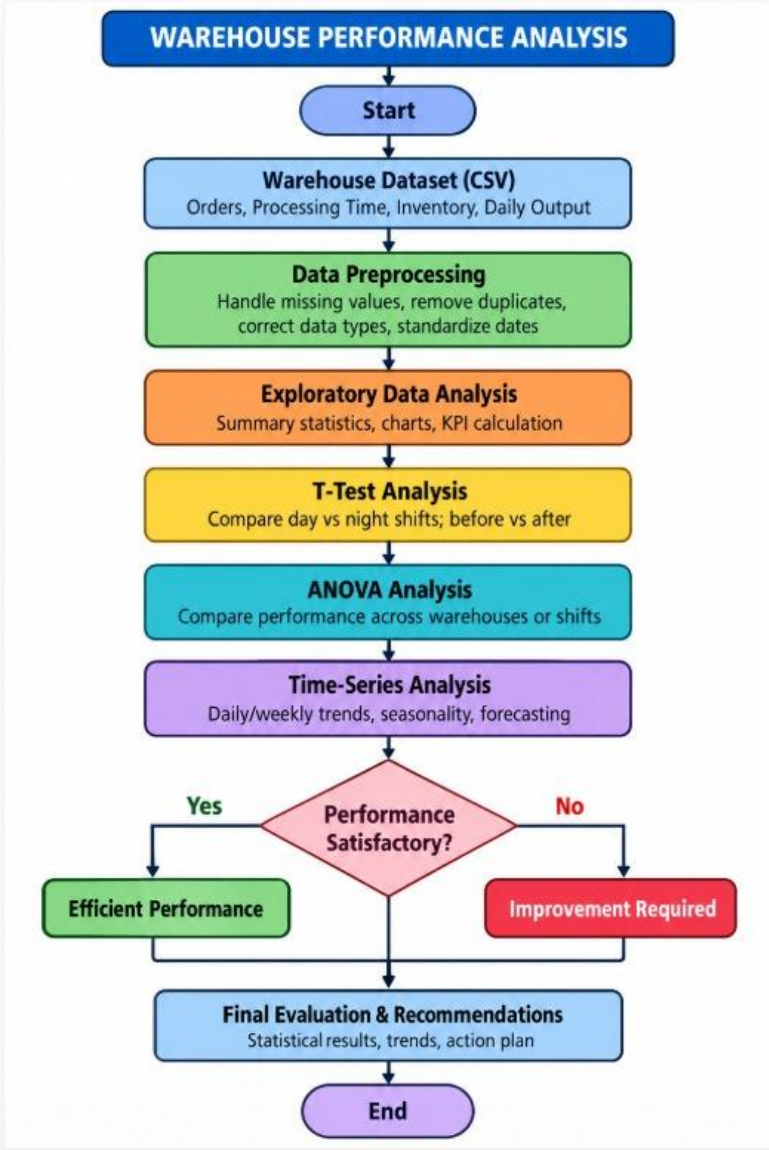


Fig 3. Integrated process flow chart

In Fig-3. The flowchart illustrates the steps involved in warehouse performance analysis, including data collection, preprocessing, exploratory data analysis, T-test, ANOVA, and time-series analysis. The final evaluation identifies warehouse performance levels and provides recommendations for improving operational efficiency.

CHAPTER-6

ALGORITHM AND RESULTS

6.1 Algorithms Used

1. T-Test – Warehouse Performance Comparison

Step-by-Step Process:

1. CType equation here.ollect warehouse performance data for two groups, such as morning and evening shifts.
2. Calculate the mean and standard deviation of each group.
3. Apply the T-test to compare the average performance of the two groups.
4. Calculate the t-statistic and p-value to determine statistical significance.
5. Interpret the results to identify differences in warehouse efficiency.

Key Equation (T-Test):

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Where:

- $\bar{x}_1, \bar{x}_2$= Mean values of two groups
- s_p= Pooled standard deviation
- n_1, n_2= Sample sizes of the two groups

2. ANOVA – Warehouse Performance Analysis

Step-by-Step Process:

1. Collect performance data from three or more warehouses or shifts.
2. Calculate the mean performance of each group and the overall mean.
3. Calculate between-group and within-group variations.
4. Compute the F-statistic and p-value to test for significant differences.
5. Identify performance variations among groups and use post-hoc tests when

necessary.

Key Equation (ANOVA F-Statistic):

$$F = \frac{MS_B}{MS_W}$$

Where:

- MS_B = Mean square between groups
- MS_W= Mean square within groups
- F= F-statistic used to evaluate group differences

3. Time-Series Analysis – Warehouse Trend Prediction

Step-by-Step Process:

1. Collect daily, weekly, or monthly warehouse data, such as order volume and processing time.
2. Arrange the data chronologically and handle missing observations.
3. Analyze trends, seasonal patterns, and fluctuations in warehouse performance.
4. Apply moving averages to smooth short-term variations.
5. Use historical trends to forecast future workload and support warehouse planning.

Key Equation (Moving Average):

$$MA_t = \frac{1}{n}\sum_{i=0}^{n-1} Y_{t-i}$$

Where:

- MA_t = Moving average at time \(t\)
- Y_{t-i} = Observed warehouse performance value
- n= Number of time periods considered

4. Data Visualization – Warehouse Performance Monitoring

Step-by-Step Process:

1. Collect the results generated by the T-test, ANOVA, and time-series analysis.
2. Use bar charts to compare warehouse or shift performance.
3. Use line graphs to display trends in orders, processing time, and delivery delays.
4. Use box plots to visualize variations in processing time.
5. Present the results in a dashboard to support performance monitoring and decision-making.

Key Equation (Average Processing Time):

$$\text{Average Processing Time} = \frac{\text{Total Processing Time}}{\text{Total Orders Processed}}$$

Where:

- Total processing time = Sum of processing durations for the orders analyzed
- Total orders processed = Number of orders included in the calculation

6.2 Result:

The proposed Warehouse Performance Analysis System was successfully implemented using statistical analysis and time-series techniques. The system processes warehouse operational data, including order processing time, inventory levels, shipment quantities, and delivery delays. The T-test was applied to compare the performance of two groups, while ANOVA was used to identify significant differences among multiple warehouses or shifts. After preprocessing, the system was tested using warehouse datasets containing daily, weekly, or monthly records.The experimental results demonstrate that the system can effectively evaluate warehouse efficiency and identify performance variations. The T-test helps determine whether differences between two groups are statistically significant, while ANOVA identifies variations among three or more groups. Time-series analysis captures trends and fluctuations in warehouse operations, helping identify changes in order volume, processing time, and delivery delays.During testing, the system generated analytical results, statistical summaries, and visualizations, including bar charts and line graphs. These outputs helped highlight inefficient processes and changing operational patterns. The analysis provides useful insights for improving resource allocation, reducing processing delays, and optimizing inventory management. The integration of all three techniques demonstrates the practical applicability of the system for warehouse performance monitoring and data-driven.

Output Images:

Missing values:

	Missing Values
Date	0
Warehouse_ID	0
Region	0
Shift	0
Orders	0
Units_Processed	0
Avg_Processing_Time_Hours	0
Picking_Accuracy_Pct	0
On_Time_Delivery_Pct	0
Inventory_Turnover	0
Labor_Hours	0
Labor_Cost	0
Operating_Cost	0
Damaged_Units	0
Stockout_Rate_Pct	0
Customer_Returns	0
Performance_Score	0

Duplicate rows: 0

Fig 4. Missing Values

In Fig-4 The dataset contains no missing values across all columns, ensuring completeness and consistency for further analysis. Additionally, no duplicate rows were detected, indicating that each record is unique. Therefore, the dataset is suitable for statistical analysis using T-test, ANOVA, and time-series techniques.

	count	mean	std	min	25%	50%	75%	max	median
Orders	21920.0	634.431524	94.885377	315.00	567.0000	636.000	703.2500	948.00	636.000
Units_Processed	21920.0	2151.549270	433.952798	664.00	1845.0000	2130.000	2439.0000	4039.00	2130.000
Avg_Processing_Time_Hours	21920.0	5.370940	0.535491	3.26	5.0000	5.370	5.7300	7.71	5.370
Picking_Accuracy_Pct	21920.0	98.102812	0.362506	96.66	97.8600	98.100	98.3500	99.55	98.100
On_Time_Delivery_Pct	21920.0	96.335173	0.664695	93.53	95.8900	96.340	96.7800	98.86	96.340
Inventory_Turnover	21920.0	8.151495	0.714913	4.95	7.6700	8.150	8.6300	11.01	8.150
Labor_Hours	21920.0	35.739726	5.282840	30.00	30.2900	34.900	39.4300	59.54	34.900
Labor_Cost	21920.0	12146.719938	1851.423090	8803.60	10524.8125	11838.925	13399.4225	20805.36	11838.925
Operating_Cost	21920.0	27368.746934	4180.631501	15728.83	24173.0300	27082.150	30257.2300	46110.98	27082.150
Damaged_Units	21920.0	10.284626	4.402277	2.00	7.0000	10.000	13.0000	29.00	10.000
Stockout_Rate_Pct	21920.0	2.586400	0.607896	0.32	2.1800	2.590	3.0000	5.11	2.590
Customer_Returns	21920.0	18.526597	6.213386	5.00	13.0000	18.000	23.0000	39.00	18.000
Performance_Score	21920.0	88.266070	0.994074	84.57	87.5900	88.280	88.9400	91.92	88.280

Fig 5.Descriptive Statistics

In Fig-5. The descriptive statistics summarize 21,920 records, showing the mean, standard deviation, minimum, maximum, and quartile values for each warehouse performance metric. The results provide insights into order volume, processing time, labor costs, delivery performance, and inventory management.

```
================================================================================
Morning mean processing time: 5.14 hours
Night mean processing time: 5.804 hours
T-statistic: -67.964
P-value: 0.0

Result:
The difference is statistically significant
at the 5% significance level.
```

	Group	Mean_Processing_Time
0	Morning	5.139608
1	Night	5.803976

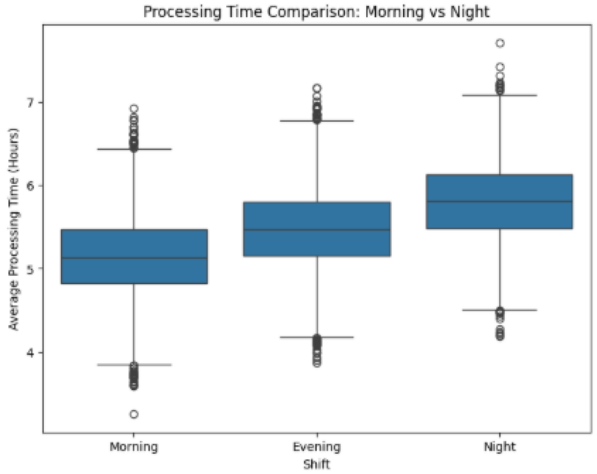


Fig 6.T-Test Morning vs Night

In Fig-6. The T-test results show a statistically significant difference in processing time between morning and night shifts, with morning averaging 5.14 hours and night averaging 5.80 hours. The box plot indicates that processing time varies across shifts, suggesting that the morning shift performs faster than the night shift.

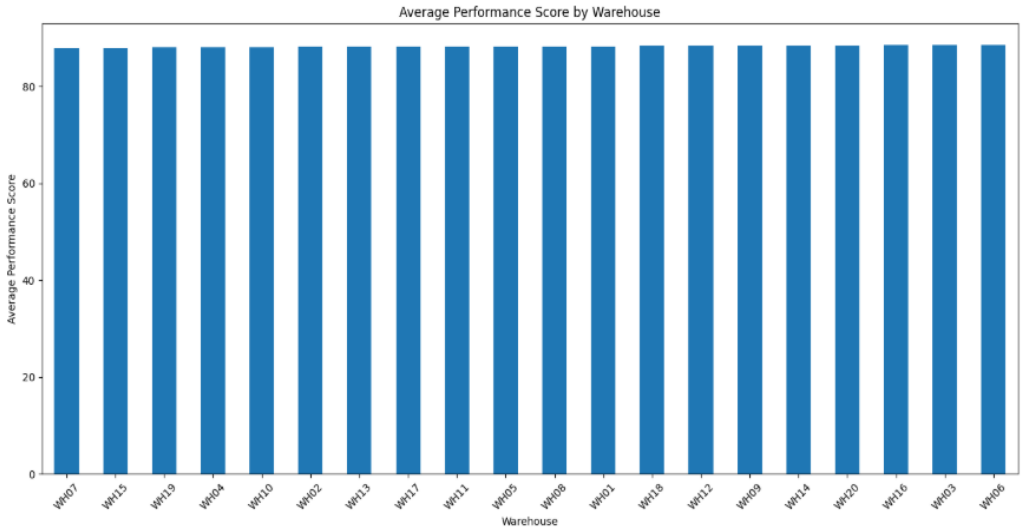


Fig 7.Average Performance by Warehouse

In Fig-7. The bar chart compares the average performance scores across different warehouses. All warehouses show similar scores of approximately 88, indicating consistent operational performance with only minor differences between warehouse locations.

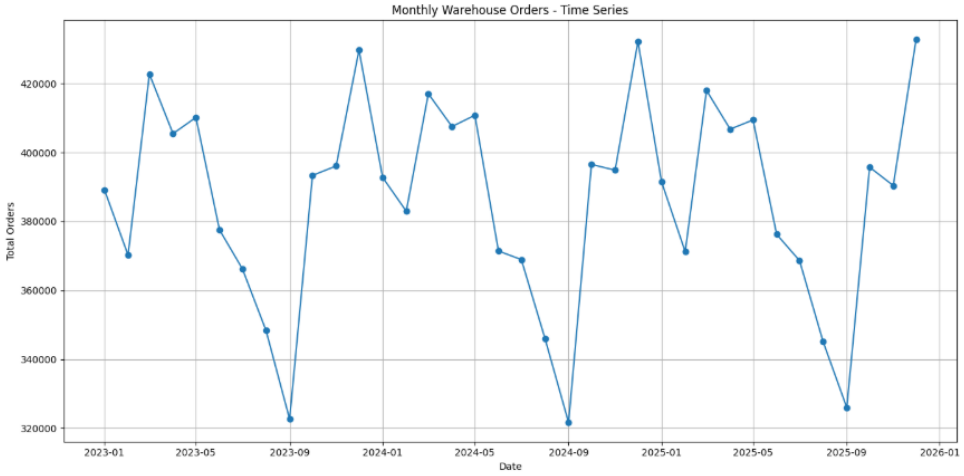


Fig 8.Monthly Warehouse Orders-Time Series

In Fig-8. The time-series graph shows monthly warehouse orders from 2023 to 2025, highlighting recurring fluctuations in order volume. Orders generally peak toward the end of the year and decline around September, indicating possible seasonal patterns that can help with future demand forecasting and resource planning.

CHAPTER-7

CONCLUSION AND FUTURE WORK

7.1.Conclusion:

The Warehouse Performance Analysis system presented in this project demonstrates an effective approach for evaluating warehouse efficiency using statistical analysis and time-series techniques. By integrating the T-test for comparing two groups, ANOVA for analyzing differences among multiple groups, and time-series analysis for identifying trends and forecasting future workloads, the system helps assess warehouse performance. Data preprocessing steps, including handling missing values, removing duplicates, and converting date formats, ensure that the input data is consistent and suitable for analysis. The system examines operational factors such as order processing time, inventory levels, shipment quantities, and delivery delays to identify performance variations and inefficiencies. It also generates visualizations, including bar charts and line graphs, to present analytical results clearly. Experimental results demonstrate how these techniques can help identify operational bottlenecks, monitor performance trends, improve resource allocation, and support data-driven decision-making for effective warehouse management.

7.2. Future Work:

Although the proposed Warehouse Performance Analysis system performs effectively, there are several opportunities for enhancement and extension. Expanding the dataset to include more warehouses, operational shifts, order volumes, and inventory records can improve the reliability of performance analysis. Incorporating advanced forecasting techniques, such as ARIMA or machine learning models, can improve the prediction of future workloads and warehouse demand. Additionally, integrating real-time data from warehouse management systems can enable continuous monitoring of operational performance and faster identification of inefficiencies. Finally, deploying the system as a web-based dashboard can improve accessibility, support real-time visualization, and help warehouse managers make better decisions regarding resource allocation, inventory management, and operational planning.

CHAPTER 8

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